

Beyond Centralized Metadata: Mitigating Cellular Re-Identification with Meshtastic Mesh Networks

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Abstract—As mobile devices become ubiquitous, the resulting cellular telemetry forms a persistent behavioral record. Modern forensic frameworks such as SURE and MoCo exploit behavioral metadata to bridge the gap between anonymity and identity, rendering traditional anonymization techniques like k-anonymity ineffective. This report analyzes these tracking frameworks and proposes the adoption of a parallel radio network based on the Meshtastic protocol as a robust mitigation. By bypassing centralized cellular operator data systems, we argue that operational security can be maintained against surveillance techniques that rely on centralized cellular metadata, as such systems operate outside the scope of the input features these models require.

Index Terms—Machine Learning, Deep Learning, Meshtastic, User Re-identification, Operational Security, LoRa.

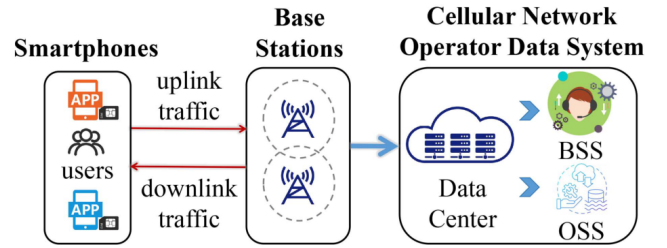
1 INTRODUCTION

1.1 The Battlefield of Data

In the modern digital landscape, data has become a highly surveilled environment where every wireless transmission leaves a trace [2]. As long as a mobile connection is maintained, a device generates a continuous stream of telemetry that forms a persistent behavioral record [2]. This footprint is captured passively by Cellular Network Operator Data Systems, which aggregate vast quantities of metadata regarding service requests, data exchanges, and handovers between tracking areas [2], [3]. Through their smartphones and the operator network infrastructure, users interact with nearby base stations for communication services such as phone calls, text messages, and Internet browsing. All of these activities are aggregated and stored in Business Support Systems (BSS) and Operating Support Systems (OSS), which serve as the foundation for operator data management [2]. Fig. 1 illustrates this data collection pipeline.

1.2 The Problem-Behavior as Identity

The fundamental challenge is that **anonymization alone does not alter behavioral patterns** [2]. Modern forensic frameworks like SURE (Smartphone Users RE-identification) and MoCo (Mobile Contact Detection) treat behavior as identity [2], [3]. By extracting behavioral signatures from sequences of Base Station (BS) associations and traffic levels, these frameworks can uniquely identify a user [2], [3]. Research indicates that the majority of user traffic is concentrated on just two antennas the “**Top 2 BS**” rule which serves as a spatial anchor for identifying a target’s home or workplace [2]. Additionally, approximately 60% of users exhibit consistently modest uplink traffic volumes, further enabling behavioral fingerprinting [2].



g. 2. Cellular network data collection.

Fig. 1. Cellular network data collection pipeline: smartphones exchange uplink/downlink traffic with base stations, which forward telemetry to the operator data center for storage in BSS and OSS databases. Adapted from [2].

1.3 The Illusion of Anonymity

Traditional security methods, such as k-anonymity and the suppression of identifiers like IMSI, are insufficient against deep learning [2], [3]. These methods fail because they only obscure identifying fields while leaving underlying spatio-temporal metadata unchanged [2], [3]. Dual-encoder architectures can mathematically bridge the gap between an anonymous index and a known identity with **AUC scores of 0.89** using only 4 days of data [2]. As shown in Fig. 2, an adversary can leverage leaked identity data and a re-identification model to de-anonymize users from publicly shared operator datasets.

1.4 Project Objectives

This report argues that remaining on a cellular grid creates a high re-identification risk [2], [3]. We propose **bypassing the cellular operator data system entirely** by deploying a parallel radio network built on the Meshtastic protocol [6].

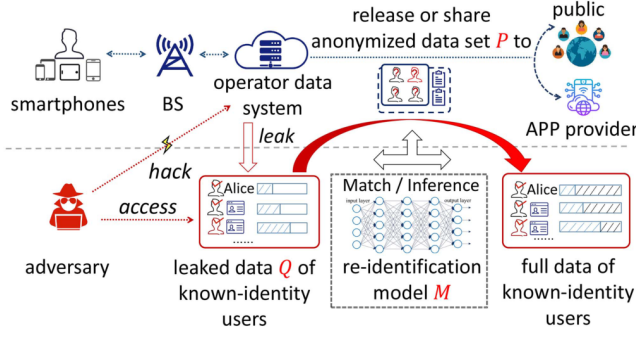


Fig. 1. Cellular user re-identification attack.

Fig. 2. Cellular user re-identification attack: an adversary uses leaked known-identity data Q and a matching model M to link anonymous records from a publicly released dataset P back to real identities. Adapted from [2].

2 BACKGROUND

2.1 Deep Learning Networks

Complex computational problems can be addressed through various methodologies, prominent among which is Deep Learning. This paradigm is fundamentally rooted in mathematical optimization. Consider a predictive model defined as:

$$\begin{aligned} f_{\theta} : \mathbb{R}^K &\rightarrow \mathbb{R}^m \\ \forall x \in \mathbb{R}^K, f_{\theta}(x) &\in \mathbb{R}^m \end{aligned} \quad (1)$$

The objective of training is to optimize the internal parameters specifically the weights and biases of the neurons to maximize the model's predictive accuracy.

This optimization process is systematically executed through the following iterative phases:

- **Initialization:** The network parameters are initially set using randomized values or predefined distribution strategies.
- **Forward Propagation:** An input vector x is processed through the mapping function f_{θ} to generate a predicted output, denoted as $\hat{y}$ (commonly referred to as the forward pass).
- **Error Evaluation:** The discrepancy between the predicted output and the ground-truth label y is quantified by evaluating a Loss function $\mathcal{L}$ (defined in Equation 2).
- **Backpropagation and Optimization:** Utilizing the backpropagation algorithm, the gradient of the loss function is computed with respect to each network weight via the chain rule. The model parameters are subsequently adjusted in the negative gradient direction a process known as gradient descent effectively minimizing the objective function.

$$\begin{aligned} \hat{y} &= f_{\theta}(x) \\ \mathcal{L} &= \mathcal{L}(\hat{y}, y) \end{aligned} \quad (2)$$

While the mathematical formulation of the loss function varies depending on the specific task architecture, its underlying objective remains constant: the resulting scalar serves

as a performance metric utilized by optimization algorithms to converge toward a local minimum¹.

Formally, Deep Learning $\subset$ Machine Learning. While both paradigms maintain distinct domains of utility, this report focuses primarily on the deployment and behavior of Deep Learning architectures.

2.2 Base Station

This infrastructure comprises cellular base stations maintained by telecommunication service providers. These stations aggregate vast quantities of telemetry and location data which, despite the application of anonymization techniques, remain vulnerable to re-identification attacks orchestrated via deep learning models. Consequently, any adversary possessing sufficient computational resources could exploit these vulnerabilities to execute large-scale mass surveillance².

2.3 The SURE Framework (User Re-identification)

The SURE (Smartphone Users RE-identification) framework [2] is a deep learning model designed to accurately classify whether two input network traces belong to the same user. It utilizes a parallel dual-encoder architecture to process anonymous and known-identity traces independently, preventing data contamination before mathematical comparison. The framework consists of four components: **Trace Encoder**, **Matching Network**, **Fusion Network**, and **Classifier**, as illustrated in Fig. 3. The Trace Encoder performs spatio-temporal embedding, which vectorizes four key privacy-impacting features: Base Station (BS) IDs, association duration, and discretized levels of uplink and downlink traffic [2]. The Matching Network then computes cross-attention weight scores between sequences of different days, capturing behavioral similarities across time. Extensive evaluations on a large-scale dataset of 10,000 users over four months (comprising more than 200 million network access records from 39,032 BSs) show that SURE achieves an AUC of 0.941, an F1-score of 0.869, and a precision of 0.807 [2]. Notably, even with a data window as narrow as four days, SURE maintains AUC scores of 0.89 with fusion features. The concentration of user activity on a small number of base stations is visualized in Fig. 4, which shows that the majority of traffic is attributed to the top two BSs per user. Further analysis reveals that older and male users exhibit higher susceptibility to re-identification risks due to more consistent behavioral patterns [2].

2.4 The MoCo Framework (Contact Detection)

MoCo (Mobile Contact Detection) [3] focuses on identifying users moving in "lockstep," such as those co-traveling in the same vehicle. The framework addresses the problem of detecting mobile contact events between urban users who exhibit similar spatio-temporal traveling behaviors, leveraging cellular signaling traces of the form $\langle \text{user ID, BS ID, timestamp} \rangle$ [3].

MoCo employs a three-stage pipeline:

1. The parameter configuration that minimizes the loss value.
2. Amnesty International: An Easy Guide to Mass Surveillance

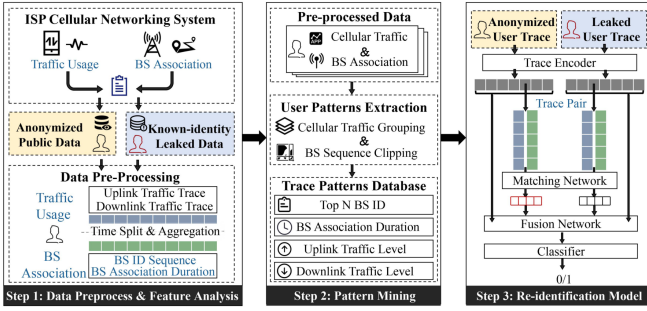


Fig. 3. Overall architecture of the SURE re-identification system.

Fig. 3. Overall architecture of the SURE re-identification system. Step 1 preprocesses raw operator data into behavioral features; Step 2 mines user patterns from traffic and BS associations; Step 3 employs a dual-encoder model (Trace Encoder, Matching Network, Fusion Network, Classifier) to determine whether two traces belong to the same user. Adapted from [2].

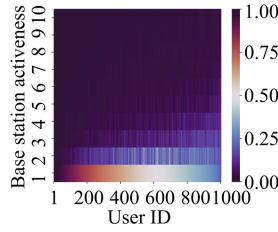


Fig. 4. Traffic usage heatmap of active BSs.

Fig. 4. Traffic usage heatmap across the top 10 most active base stations for 1,000 users. The bottom two rows (BS 1 and BS 2) concentrate the vast majority of each user's traffic, illustrating the "Top 2 BS" rule that enables spatial fingerprinting. Adapted from [2].

- **Data Denoising:** The DBSCAN clustering algorithm is applied to remove noise from remote BS connections that do not reflect actual user locations [3].
- **Spatio-Temporal Filter (ST-Filter):** This module eliminates unlikely mobile contact trajectory pairs in both spatial and temporal domains, typically requiring $\geq 60\%$ spatial similarity to retain candidate pairs [3].
- **Detection Learning Network:** An LSTM-based spatial branch captures correlations among candidate trajectories using quadkey-encoded BS sequences, while an MLP-based temporal branch learns time-step-dependent patterns from one-hot encoded trajectory vectors [3].

The model is trained using cross-entropy loss and outputs a binary mobile contact classification. Extensive experiments on real-world data (241 user trajectories spanning 211 hours and 6,204 km) demonstrate that MoCo achieves an average F1-score of 87.13%, significantly outperforming classical methods such as S-LCSS and ST-LCSS, as well as deep learning baselines including DPLink and DeepTUL [3]. Fig. 5 provides an overview of MoCo's three-stage design.

2.5 Supply Chain Vulnerabilities in Learning-Based Pipelines

Although SURE and MoCo rely on supervised learning, related work in self-supervised learning (SSL) reveals broader

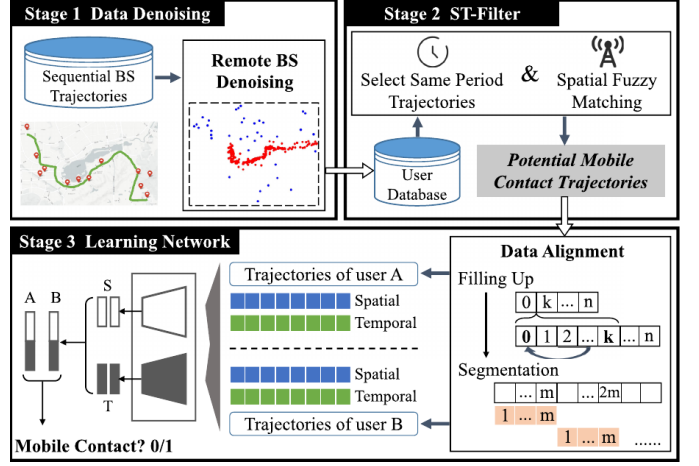


Fig. 5. Design overview of MoCo. Stage 1 applies DBSCAN-based denoising to remove remote BS noise; Stage 2 filters trajectory pairs using spatio-temporal similarity; Stage 3 uses a combined LSTM (spatial) and MLP (temporal) learning network with data alignment to classify mobile contact events. Adapted from [3].

vulnerabilities that are relevant to centralized data pipelines. Studies on backdoor attacks show that an adversary can poison as little as 0.5% of unlabeled training data with a rigid trigger patch, causing an SSL model to associate that trigger with a target category at test time [7], [8]. Recent work such as DIFT [8] proposes defenses against data poisoning in contrastive learning, while research on adversarial susceptibility [4] demonstrates that contrastive self-supervised models exhibit higher adversarial susceptibility than their supervised counterparts. These findings illustrate that learning-based pipelines introduce additional attack surfaces beyond metadata privacy, though this represents a different threat vector than the re-identification exploited by SURE and MoCo.

3 PROPOSED APPROACH: MESHTASTIC PARALLEL NETWORK

3.1 Technical Architecture

We propose **Meshtastic**, a decentralized mesh network utilizing the **LoRa physical layer** and Chirp Spread Spectrum (CSS) [6]. Signals consist of frequency-sweeping "chirps" with a 16-up-chirp Preamble for synchronization and a specific Sync Word (0x2B) to distinguish the network [6].

3.2 Decentralized Metadata Protection

The primary advantage is the complete bypass of Cellular Operator Data Systems [6]. In a mesh architecture, there is no central database logging "Top 2 BS" anchors or hourly traffic windows the features required by SURE and MoCo for re-identification [2], [3], [6]. The SURE framework relies on extracting BS ID sequences, association durations, and traffic levels from centralized operator BSS/OSS databases [2]. Similarly, MoCo requires access to cellular signaling traces of the form (user ID, BS ID, timestamp) stored by operators [3]. By removing the centralized collection point entirely, Meshtastic eliminates the cellular operator metadata source that makes these attacks possible.

3.3 Encryption and Routing

Meshtastic payloads are encrypted using **AES-256-CTR encryption** over LoRa, with a different key for each channel [6]. Only nodes with the correct channel key can interpret data, while hop limits constrain message propagation, reducing the volume of accessible metadata compared to centralized operator logging [6].

3.4 Justification of Selection

The choice of a parallel radio network over software-side mitigations is motivated by the fundamental limitations of the latter. Approaches such as noise injection and differential privacy attempt to obscure behavioral patterns at the data level; however, they suffer from an inherent trade-off: sufficient noise to defeat re-identification may render the data unusable for its intended purpose. The SURE paper demonstrates that even with k-anonymity (identifier suppression) applied, AUC scores remain above 0.9, indicating that anonymization alone is insufficient [2]. Moreover, techniques like ASTRA [1] and RoCL [5] harden the model against adversarial perturbations but do not address the root cause: **the centralized logging of spatio-temporal metadata by cellular operators**. Meshtastic addresses this root cause by operating on an entirely separate radio infrastructure, ensuring that no cellular operator ever collects the behavioral traces in the first place.

4 SWOT ANALYSIS

To rigorously evaluate the position of the proposed Meshtastic parallel network against existing cellular infrastructure, we conduct a SWOT (Strengths, Weaknesses, Opportunities, Threats) analysis. This comparative framework establishes the credibility of our proposal by systematically examining both internal attributes and external factors.

TABLE 1
SWOT Analysis: Meshtastic Parallel Network vs. Cellular

	Internal	External
S/O	No centralized operator logs; metadata privacy; AES-256-CTR encryption [6].	Growing demand for off-grid, high-privacy communication solutions.
W/T	Lower data rates; limited LoRa range; no PFS [6].	Signal triangulation; RF fingerprinting; LoRa band jamming.

Strengths. The most significant internal advantage is the complete absence of a centralized operator data system. As demonstrated by SURE [2], re-identification attacks require access to BS association sequences and traffic usage patterns stored in BSS/OSS databases. By removing this centralized collection point, Meshtastic eliminates the “Top 2 BS” spatial anchors and hourly traffic windows that enable behavioral fingerprinting. Additionally, AES-256-CTR shared-key encryption ensures payload confidentiality at the application layer [6].

Weaknesses. LoRa operates in the sub-GHz ISM bands with data rates limited to approximately 5.47 kbps at the highest spreading factor, making it unsuitable for high-bandwidth applications such as video streaming or large file transfers [6]. The propagation range, while superior to WiFi in line-of-sight conditions, is constrained in dense urban environments with significant signal attenuation. Furthermore, the mesh topology introduces latency with each additional hop. From a security standpoint, Meshtastic does not provide Perfect Forward Secrecy, message integrity verification, or sender authentication [6], meaning that an attacker with access to the channel key can impersonate any node on the network and decrypt passively collected traffic. In the threat model addressed by this report, however, these cryptographic deficiencies are secondary concerns: the primary objective is to eliminate the centralized cellular metadata stream that enables mass behavioral profiling. An attacker who compromises a Meshtastic channel key can surveil at most the traffic of that single mesh group, whereas a compromised cellular operator database exposes the behavioral records of millions of users.

Opportunities. The increasing demand for operational security in privacy-sensitive domains creates a growing market for off-grid communication solutions. Recent data breaches at major operators (AT&T, T-Mobile, US Cellular) [2] have highlighted the vulnerability of centralized data repositories, strengthening the case for decentralized alternatives. Application domains such as journalism, activism, and disaster relief stand to benefit from infrastructure-independent communication.

Threats. While Meshtastic avoids cellular metadata collection, it remains potentially vulnerable to RF-level attacks. Signal triangulation using directional antennas could determine the physical location of transmitting nodes. Future advances in RF fingerprinting which identify unique hardware characteristics of individual radio transmitters could theoretically enable device-level identification without requiring centralized metadata logs. Additionally, intentional signal jamming in the LoRa frequency bands could disrupt mesh network operations.

5 EVALUATION

5.1 In-Scope Assessment of Re-Identification Attacks

A rigorous evaluation of the proposed Meshtastic parallel network against the SURE and MoCo frameworks requires first examining the precise input requirements of each attack model. Fundamental to this assessment is the concept of **scope**: a model can only operate on data that falls within its representational domain.

5.2 Input Feature Incompatibility

The SURE framework [2] requires four privacy-impacting input features extracted from cellular operator BSS/OSS databases: Base Station (BS) IDs, association duration, and discretized levels of uplink and downlink traffic. These features are derived from the continuous telemetry stream generated when a mobile device maintains a connection with the cellular network. Similarly, MoCo [3] requires cellular signaling traces of the form (user ID, BS ID, timestamp) to detect co-traveling patterns.

Since the Meshtastic protocol operates on an entirely independent LoRa-based radio infrastructure, it generates none of these data types. There is no concept of a “base station” in a mesh network only peer nodes. There is no centralized operator collecting uplink/downlink traffic volumes, no BS association logs, and no signaling trace database. The input feature space that SURE and MoCo operate over simply does not exist in a Meshtastic deployment.

5.3 Scope Boundary Argument

Because the SURE and MoCo models are defined exclusively over cellular operator metadata, a Meshtastic user falls entirely outside their representational scope. This is not a hypothesis to be tested empirically but a direct consequence of the input feature definitions:

- **SURE:** Input = BS ID sequence + association duration + uplink traffic + downlink traffic. Meshtastic provides none of these. Therefore SURE cannot form a valid input vector.
- **MoCo:** Input = $\langle \text{user ID, BS ID, timestamp} \rangle$ traces. Meshtastic does not associate users with BSs. Therefore MoCo cannot construct candidate trajectory pairs.

By virtue of operating outside the domain of the input features these models require, Meshtastic users are not susceptible to the re-identification and contact-detection attacks described by SURE and MoCo as currently designed. This conclusion follows from the definitional mismatch between the data each attack requires and the data a Meshtastic deployment produces. We note, however, that this does not preclude the development of future models for instance, a LoRa-aware variant of SURE that could exploit Meshtastic metadata such as node IDs, hop counts, or signal strength records should such data become available to an adversary.

5.4 Acknowledged Residual Risks

While the cellular metadata channel is eliminated, Meshtastic users remain potentially vulnerable to RF-level attacks outside the scope of SURE and MoCo. Signal triangulation via directional antennas can localize transmitting nodes, and emerging RF fingerprinting techniques could theoretically identify individual LoRa transmitters based on hardware-specific signal characteristics. Additionally, nodes periodically broadcast position, telemetry, and node-info updates over the primary channel [6], which could be intercepted by any receiver within range. These risks are not captured by the SURE or MoCo frameworks but warrant acknowledgment as orthogonal threat vectors.

6 RELATED WORK

6.1 The Collapse of K-Anonymity

Traditional anonymization relies on the removal or suppression of direct identifiers such as IMSI numbers, phone numbers, and subscriber names. However, research has demonstrated that these techniques are fundamentally insufficient when spatial behavior remains stable [2]. The SURE framework shows that even with k-anonymity applied to shared datasets, the underlying spatio-temporal

metadata BS associations, traffic patterns, and temporal routines persists unchanged, enabling re-identification with AUC scores exceeding 0.9 [2].

A critical finding from the SURE study is that user demographics significantly impact re-identification susceptibility. Male users and those aged 45 and above exhibit the highest re-identification risks (AUC scores of 0.95 for males and above 0.93 for users over 45), attributable to their more consistent and predictable logistical routines [2]. Conversely, users aged 18–24 show marginally lower risk, likely due to more variable mobility patterns. These findings demonstrate that the “Top 2 BS” spatial anchor, where the majority of a user’s traffic concentrates on just two base stations, provides a near-deterministic fingerprint of home and workplace locations that survives any identifier-level anonymization [2].

Furthermore, MoCo [3] extends the threat beyond individual re-identification to social graph reconstruction. By detecting co-traveling patterns with an F1-score of 87.13%, MoCo can map a target’s “inner circle” the set of users who regularly share transportation routes using only cellular signaling traces [3]. This compound threat renders k-anonymity not only insufficient for identity protection but also incapable of shielding social associations.

6.2 Alternative SSL Defenses

Several recent works have proposed defensive techniques against adversarial attacks on self-supervised learning (SSL) models. **ASTRA** (Adversarial Self-Supervised Training with Adaptive-Attacks) [1] introduces an adaptive adversarial training procedure that generates perturbations during the pre-training phase, improving robustness against adversarial inputs. Similarly, **RoCL** (Robust Contrastive Learning) [5] incorporates adversarial examples directly into the contrastive learning objective, training the encoder to produce consistent representations even under input perturbations.

While these defenses are effective against model-level adversarial attacks, they do not address the root cause of the tracking threat analyzed in this report: **the centralized logging of spatio-temporal metadata by cellular operators** [2], [3]. ASTRA and RoCL harden the learning pipeline but leave the data collection infrastructure untouched. An adversary with access to operator BSS/OSS databases can simply train new models on the same raw metadata, bypassing any model-side defense. Additionally, DIFT [8] proposes density-based identification and fine-tuning against data poisoning backdoor attacks in contrastive learning, but this again operates at the model level rather than addressing data collection.

This fundamental limitation motivates our proposed approach: rather than defending the model, we eliminate the centralized data source by deploying a parallel communication network that never generates the metadata exploited by SURE and MoCo.

7 CONCLUSIONS

7.1 Summary of Findings

As long as cellular connections are maintained, **a high re-identification risk** persists due to behavioral predictability

[2], [3]. The analysis presented in this report leads to the following key conclusions:

- **Behavioral metadata can function as a highly distinctive identifier.** The SURE framework demonstrates that BS association patterns and traffic usage create unique behavioral fingerprints, achieving re-identification AUC scores of 0.941 on datasets of 10,000 users, even with k-anonymity applied [2].
- **Social graphs are exposed.** MoCo extends the threat from individual identification to social network reconstruction, detecting co-traveling contacts with an F1-score of 87.13% using only cellular signaling traces [3].
- **Software mitigations are insufficient.** Traditional anonymization techniques and model-hardening approaches fail to address the root cause: centralized metadata collection by cellular operators. While defenses such as ASTRA [1] and RoCL [5] improve model robustness, they operate at the learning level and leave the underlying data pipeline untouched.
- **Physical-layer separation offers a compelling alternative.** The Meshtastic parallel radio network eliminates the cellular operator metadata source exploited by both SURE and MoCo by operating on a completely independent infrastructure with no centralized logging [6].

Parallel radio networks like Meshtastic offer a pathway for operating outside the centralized cellular data ecosystem [6].

7.2 Future Extensions

Several directions for future work emerge from this analysis:

- **Multi-hop scalability.** Investigating the performance of Meshtastic mesh networks at scale, particularly the impact of increasing hop counts on latency, reliability, and metadata leakage through traffic analysis.
- **Hybrid encrypted mesh systems.** Combining Meshtastic's LoRa backbone with short-range protocols (e.g., BLE mesh) to increase bandwidth for local communications while maintaining the metadata-free properties of the long-range network.
- **RF fingerprinting resistance.** Developing countermeasures against emerging RF fingerprinting techniques that could potentially identify individual LoRa transmitters based on hardware-specific signal characteristics.
- **Experimental validation.** Conducting the proposed evaluation framework with real user groups to empirically quantify the re-identification risk reduction achieved by Meshtastic compared to cellular base-lines.

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